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**NITTE (Deemed to be University)**

(Established under Sec.(3) of UGC Act, 1956)

**First Semester (CBCS) Degree Examinations, NOV. - DEC. 2025****25CHY102 – CHEMISTRY FOR CIVIL ENGINEERING**

(For CIV)

Duration: 3 Hours]

[Max. Marks: 100

**Note:**

**Part – A:** Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B:** Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

- Brass is an alloy of .....  
A) Iron, copper and zinc  
B) Iron and copper  
C) Iron and zinc  
D) Copper and zinc
- Which of the following statements is NOT true for iron metal?  
A) It is malleable and ductile  
B) It is good conductor of electricity  
C) It is highly corrosion resistant  
D) It is magnetic
- Which type of ceramic is commonly used in spark plugs?  
A) Porcelain  
B) Earthenware  
C) Stoneware  
D) Glass-ceramic
- Annealing of glass is carried out to .....  
A) Remove internal stresses  
B) Improve color  
C) Increase brittleness  
D) Increase porosity
- What is the electrode potential of the calomel electrode with saturated KCl solution at 298K?  
A) 0.334 V  
B) 0.281 V  
C) 0.242 V  
D) 0.0591 V
- Which of the following is an example for ion selective electrode?  
A) Platinum electrode  
B) Gas electrode  
C) Calomel electrode  
D) Glass electrode
- Photovoltaic cells convert ..... energy to ..... energy.  
A) Chemical; Electrical  
B) Electrical; Chemical  
C) Solar; Electrical  
D) Electrical; Solar
- Which of the following is not a feature of secondary batteries?  
A) Rechargeable  
B) Long-lasting when charged  
C) Non-rechargeable  
D) Can be used after recharging
- Complete the statement: In differential aeration corrosion, .....  
A) Oxygen-deficient part gets protected.  
B) Oxygen-deficient part acts as anode.  
C) Oxygen-rich part acts as anode.  
D) Metal as a whole acts as cathode.
- Ships sailing in ocean suffer from .....  
A) Stress corrosion  
B) Grain boundary corrosion  
C) Pitting corrosion  
D) Water line corrosion
- The metal, which is protected by a layer of its own oxide, is .....  
A) Al  
B) Pt  
C) Zn  
D) Fe
- In general, corrosion is maximum when the pH of the corroding medium is .....  
A) Above 7.0  
B) Equal to 7.0  
C) Below 7.0  
D) Equal to 10.0

13. Common raw materials for geopolymer synthesis are .....  
 A) Limestone and clay B) Fly ash and slag  
 C) PVC and polyethylene D) Epoxy and nylon
14. Polylactic acid (PLA) is synthesized from .....  
 A) Lactide (cyclic dimer of lactic acid) B) Ethylene glycol  
 C) Styrene D) Caprolactam
15. The functional group responsible for epoxy resin properties is .....  
 A) Hydroxyl group B) Amide group  
 C) Epoxide ring D) Ester group
16. Which molecule is eliminated during the condensation polymerization of Nylon-6,6?  
 A) Water B) Methanol  
 C) Nitrogen D) Carbon dioxide
17. Top-down approach of nanomaterial synthesis involves .....  
 A) Self-assembly B) Lithography  
 C) Sol-gel method D) Precipitation
18. Potentiometric titrations are based on .....  
 A) Sudden change in EMF B) Current flow  
 C) Resistance D) Pressure change
19. Conductometric titration can be applied to .....  
 A) Mixture of acids B) Non-electrolytes  
 C) Mixture of solids D) Metals only
20. Which factor does NOT affect Beer's law?  
 A) Concentration B) Path length  
 C) Pressure D) Wavelength

## PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

### Unit – I

|                                                                                                                                               | Marks | BT* | CO* | PO* |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) Highlighting the principle, describe experimental procedure to estimate the amount of calcium oxide in cement by complexometric method. | 7     | 2   | 1   | 1   |
| b) Define refractory. Classify refractories based on the chemical constituents and give an example for each.                                  | 4     | 1   | 1   | 1   |
| c) Mention any five properties of glass.                                                                                                      | 5     | 1   | 1   | 1   |
| 2. a) Differentiate Duralumin and Magnalium based on their composition and properties. List two applications of each.                         | 8     | 3   | 1   | 1   |
| b) With suitable sketches, outline the steps involved in the preparation of soda-lime glass.                                                  | 8     | 2   | 1   | 1   |

### Unit – II

|                                                                                                                                                                                                 |   |   |   |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 3. a) Explain the construction and working of calomel electrode. Mention its advantages.                                                                                                        | 8 | 2 | 2 | 1 |
| b) Define concentration cell. For the concentration cell, $\text{Ni(s)}   \text{NiCl}_2 (2 \text{ M})    \text{NiCl}_2 (64 \text{ M})   \text{Ni(s)}$ , calculate the EMF of the cell at 298 K. | 4 | 3 | 2 | 1 |
| c) Enumerate four advantages of power alcohol.                                                                                                                                                  | 4 | 1 | 2 | 1 |
| 4. a) Why is glass electrode called as "pH electrode"? Explain its construction.                                                                                                                | 4 | 2 | 2 | 1 |
| b) Discuss construction of lithium-ion battery. Explain the charging and discharging process by giving suitable reactions.                                                                      | 8 | 2 | 2 | 1 |
| c) Write the cathodic and anodic reactions during water electrolysis. Mention any two advantages of green hydrogen as a fuel.                                                                   | 4 | 2 | 2 | 1 |

### Unit – III

|    |    |                                                                                                                       |   |   |   |   |
|----|----|-----------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 5. | a) | Explain the electrochemical theory of corrosion taking Fe as an example.                                              | 8 | 2 | 3 | 1 |
|    | b) | Discuss how the following factors affect the rate of corrosion:<br>(i) Hydrogen overvoltage (ii) Humidity             | 4 | 2 | 3 | 1 |
|    | c) | "Cathodic inhibitors are used to control corrosion" Justify.                                                          | 4 | 1 | 3 | 1 |
| 6. | a) | Outline and explain the concept of differential aeration corrosion by considering waterline and pitting corrosion.    | 7 | 2 | 3 | 1 |
|    | b) | List out the factors to be considered while selecting materials for corrosion prevention in engineering applications. | 4 | 1 | 3 | 1 |
|    | c) | Explain the steps involved during the galvanization of iron.                                                          | 5 | 2 | 3 | 1 |

### Unit – IV

|    |    |                                                                                                                                                                                                               |   |   |   |   |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 7. | a) | Classify polymers based on tacticity.                                                                                                                                                                         | 3 | 1 | 4 | 1 |
|    | b) | Define glass transition temperature ( $T_g$ ). How the following factors affect $T_g$ .<br>(i) Flexibility (ii) Crystallinity (iii) Addition of plasticizers                                                  | 7 | 2 | 4 | 1 |
|    | c) | Give the synthesis of Kevlar. Mention its salient properties and applications.                                                                                                                                | 6 | 2 | 4 | 1 |
| 8. | a) | A polymer sample contains 50% molecules having molecular weight of 5000, 40% molecules having molecular weight of 4000 and 10% molecules having molecular weight of 1000. Calculate its polydispersity index. | 5 | 3 | 4 | 1 |
|    | b) | Explain the synthesis of Nylon-6,6. How does it differ from natural fibres in terms of its properties?                                                                                                        | 7 | 2 | 4 | 1 |
|    | c) | Discuss the process of Geopolymer concrete synthesis using fly ash as a source.                                                                                                                               | 4 | 2 | 4 | 1 |

### Unit – V

|     |    |                                                                                                                                                |   |   |   |   |
|-----|----|------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| 9.  | a) | Write the Henderson's equation and apply the equation to determine the pKa of acetic acid. Justify your answer by plotting appropriate graphs. | 7 | 3 | 5 | 1 |
|     | b) | List out the advantages of instrumental analysis over chemical analysis.                                                                       | 4 | 1 | 5 | 1 |
|     | c) | Explain the synthesis of carbon nanotubes through chemical vapor deposition method.                                                            | 5 | 2 | 5 | 1 |
| 10. | a) | Discuss, how does the conductance of HCl vary when neutralized with NaOH? Explain the variation in conductance and plot the graph.             | 4 | 2 | 5 | 1 |
|     | b) | Classify nanomaterials based on their dimensions.                                                                                              | 4 | 2 | 5 | 1 |
|     | c) | Describe the sol-gel method of synthesizing silica aerogel.                                                                                    | 8 | 2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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